

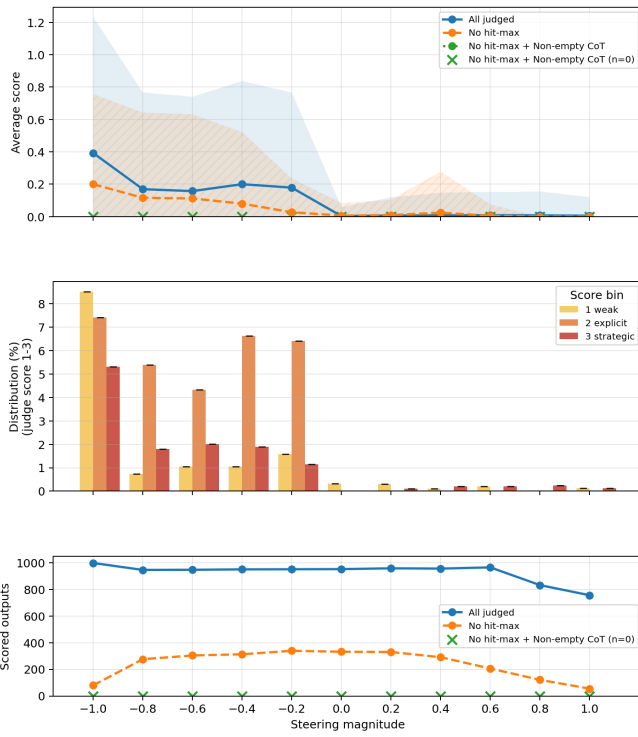


Figure 18: Best-layer results across steering strengths α for Olmo3 7B Base.

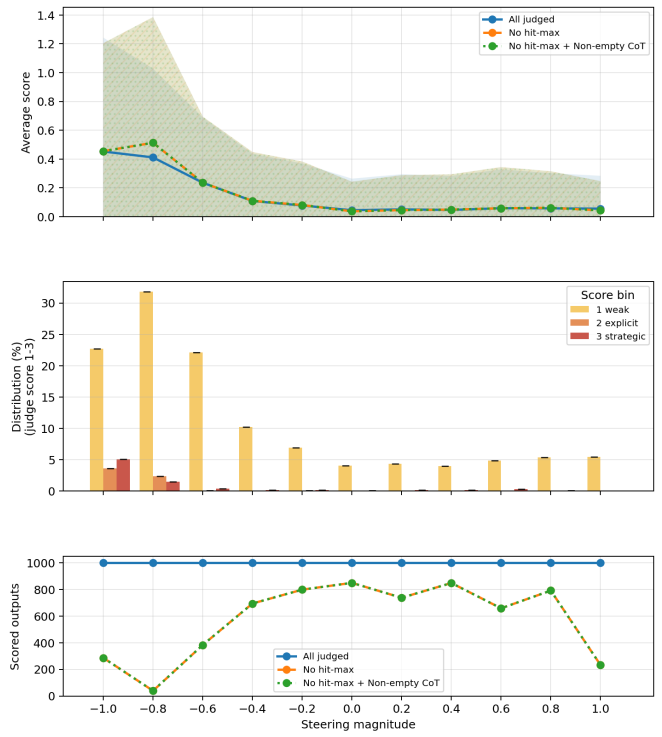


Figure 19: Best-layer results across steering strengths α for Olmo3 7B SFT.

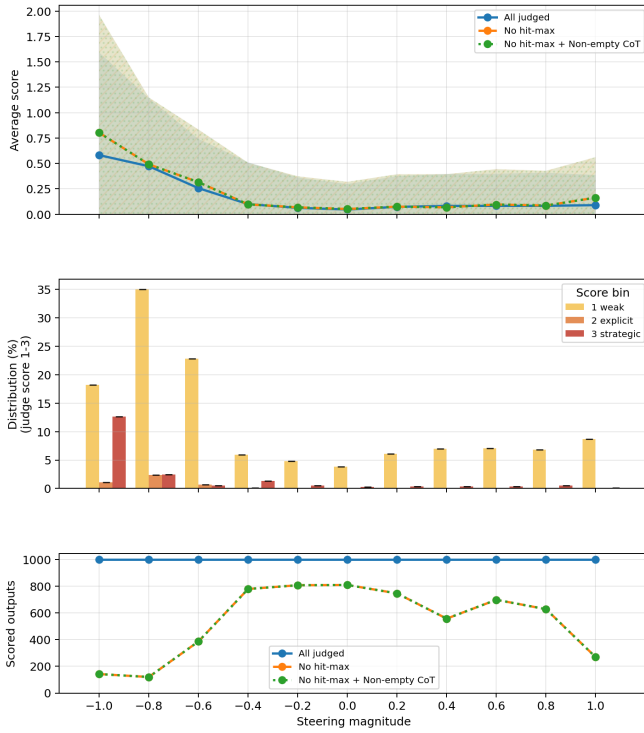


Figure 20: Best-layer results across steering strengths α for Olmo3 7B DPO.

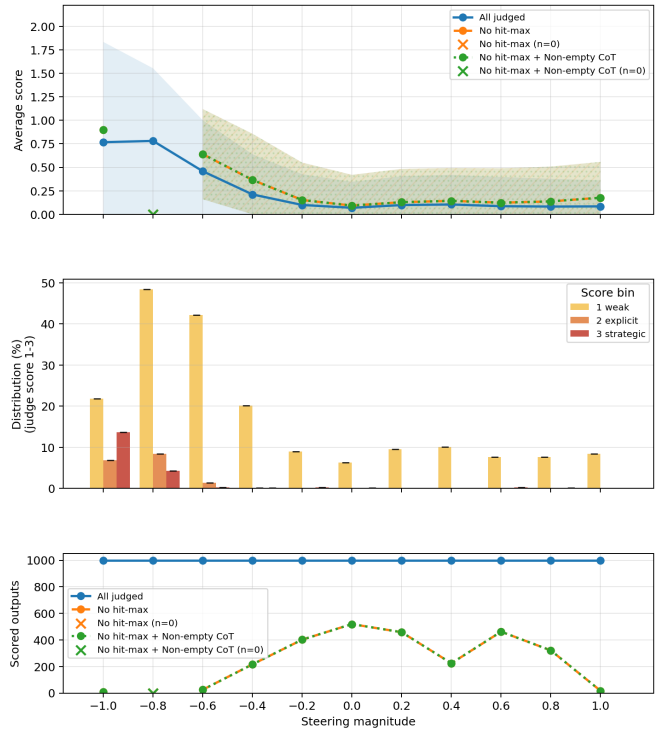


Figure 21: Best-layer results across steering strengths α for Olmo3 7B Think.

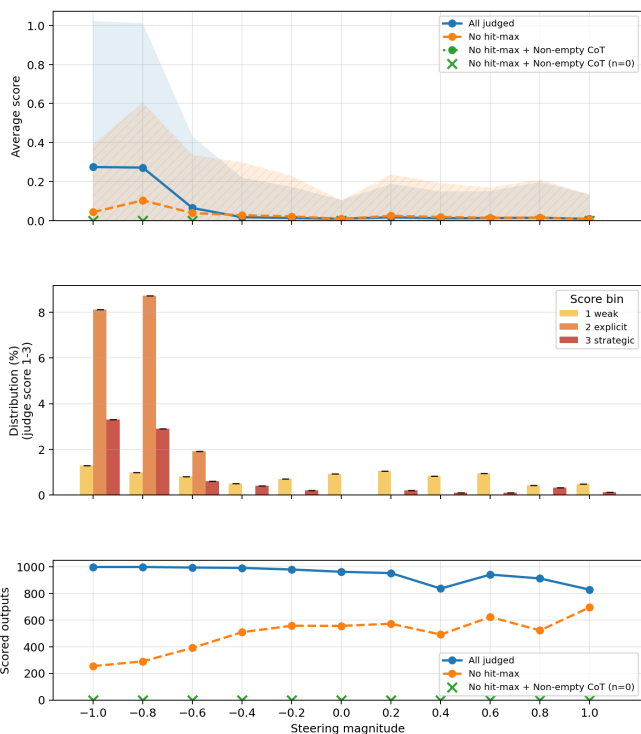


Figure 22: Best-layer results across steering strengths α for Olmo3 32B Base.

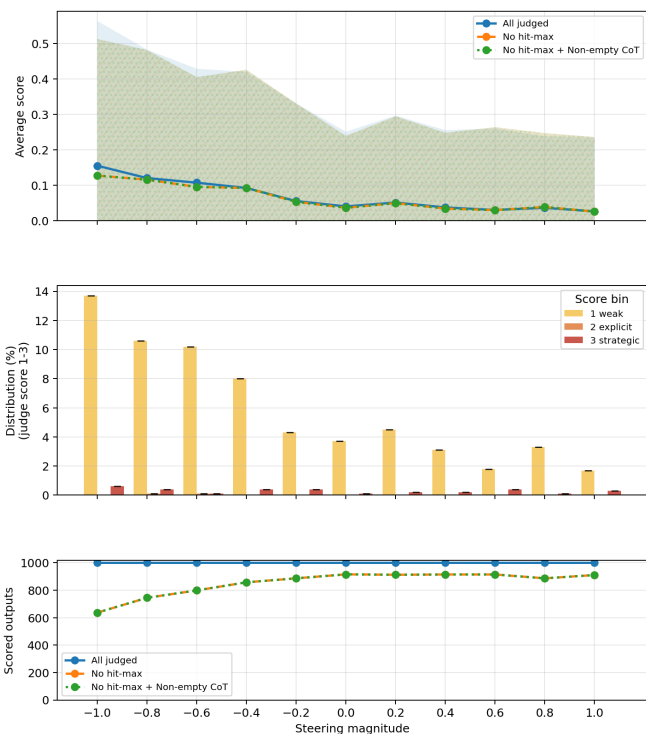


Figure 23: Best-layer results across steering strengths α for Olmo3 32B SFT.

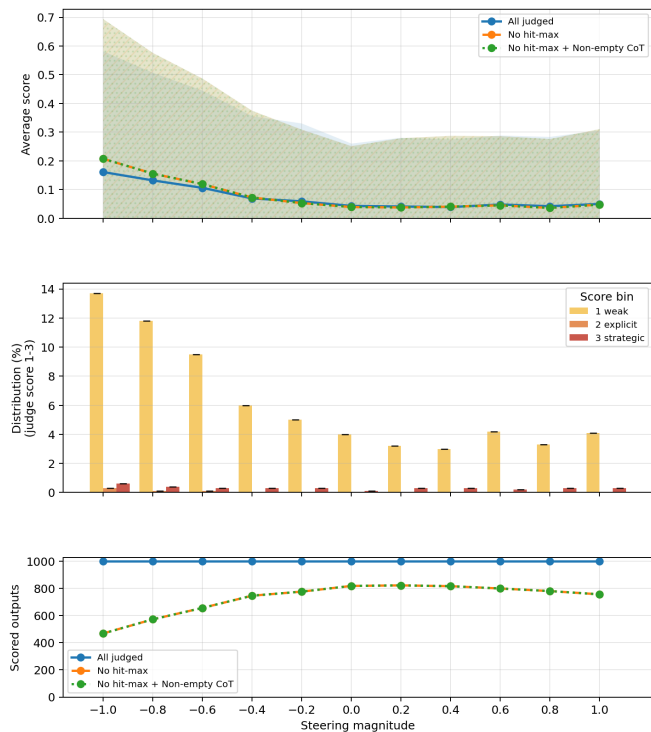


Figure 24: Best-layer results across steering strengths α for Olmo3 32B DPO.

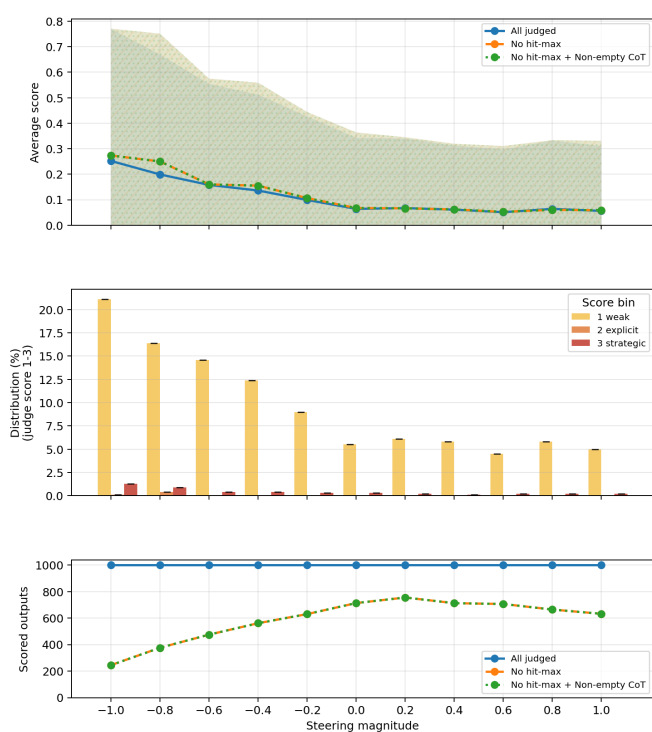


Figure 25: Best-layer results across steering strengths α for Olmo3 32B Think.

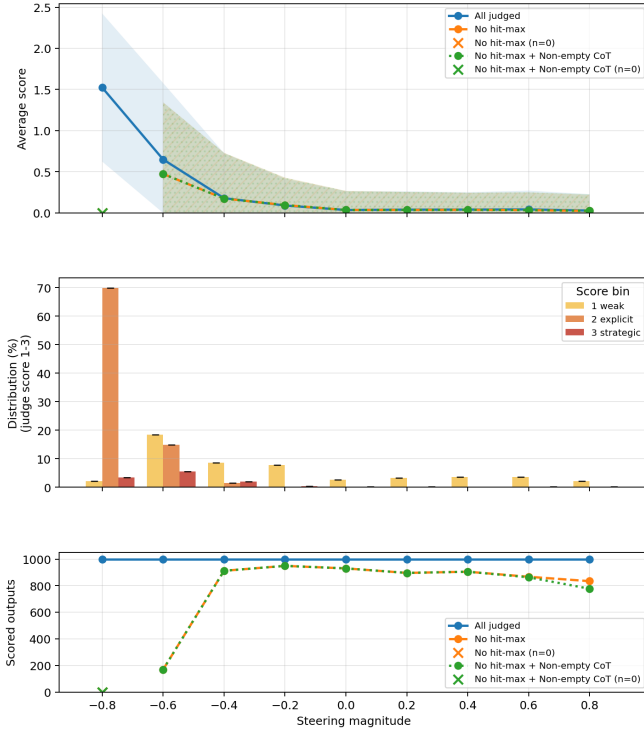


Figure 26: Best-layer results across steering strengths α for Qwen3 8B.

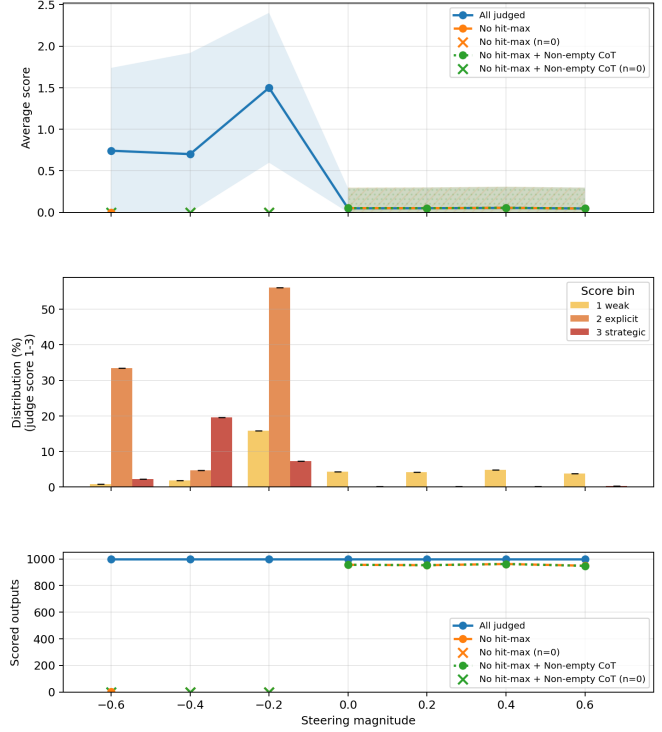


Figure 27: Best-layer results across steering strengths α for Qwen3 32B.

I.1 Probe Direction Cosine Similarity Across Stages

Figure 28 reports pairwise cosine similarity of the probe direction $\hat{v}_\ell$ across Olmo training stages, at the model-best layer. SFT, DPO, and Think share a probe direction at cosine ≥ 0.998 for both 7B and 32B; the Base direction is distinct (cosine 0.72 at 7B and 0.83 at 32B against the post-trained directions). Post-training therefore stabilizes a probe direction already present at Base rather than producing a new direction at each stage.

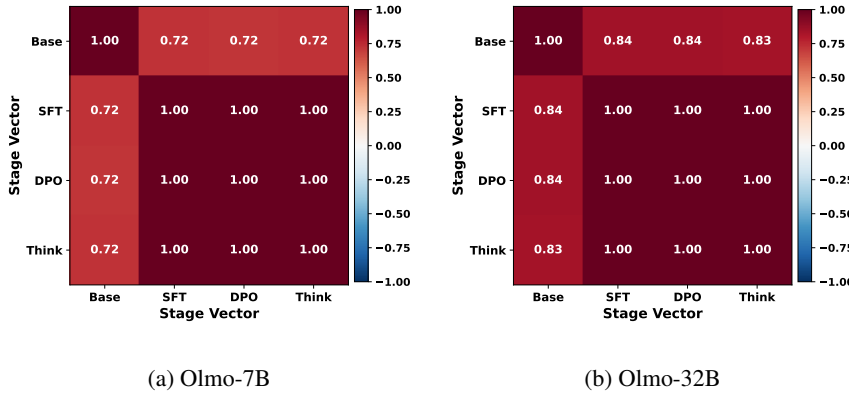


Figure 28: Cosine similarity matrices of the probe direction $\hat{v}_\ell$ across training stages. SFT, DPO, and Think share a direction at cosine ≥ 0.998 in both models; Base is markedly distinct.

I.2 Per-prompt Coupling Across Training Stages

Figure 29 reports per-prompt coupling at the 7B scale across the four Olmo training stages, and Figure 30 reports the corresponding 32B results. Both scales show the same Base $\rightarrow$ Think trend: peak $|\rho_\ell|$ at Base does not exceed 0.05 and peak $|\rho_\ell|$ at Think does not exceed 0.15, and peak MI at Base sits at the random-direction baseline while SFT, DPO, and Think exceed it by approximately 0.005–0.015 nats. The Base $\rightarrow$ Think change in prompt-wise coupling is therefore present but small in absolute terms, and smaller than the corresponding AUROC change in Figure 3. This shows that post-training amplifies the linearly-decodable signal more than it increases its coupling (measured by correlation and mutual-information) to verbalization.